

WHEN ARE EXACT CONSERVATION LAYERS PLUG-AND-PLAY?

IDENTIFYING TRAINING–INFERENCE PATH DEPENDENCE IN NEURAL PDE SURROGATES

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ABSTRACT

Projected training can make a conservation layer part of the learned autoregressive transition. Removing the layer then changes conserving dynamics through feedback. We distinguish output coordinates, projection during training, and projection during inference with an eight-cell intervention design. Projected training identifies the deployed predictor only up to an unconstrained raw invariant component. In a prospectively frozen, 30-seed PDEBench Advection study, the three-way interaction is 0.22974 [0.19644, 0.26345], material in 9/12 mandatory resolution–horizon cells. Opposite-signed training and inference credits obscure this dependence in the ordinary bundled comparison; removal damage emerges through autoregressive feedback. An independently frozen, 30-seed shallow-water study reproduces material path dependence in 6/8 cells. A direct input intervention also establishes that changes along the invariant direction can affect conserving dynamics in the tested FNOs. The registered U-Net transfer has an unresolved primary interaction and only 3/12 material cells; post-hoc replay places its eight native-grid primary cells below persistence. A separate post-hoc U-Net deployment control, computed on a fixed spatial grid, is predictive and exhibits path dependence, but changes the deployment strategy and does not replace that registered transfer. These results show that, in the tested FNOs, the conservation layer participates in the learned autoregressive transition. Deployment changes that bypass it therefore require validation of the resulting rollout dynamics.

1 INTRODUCTION

Learned simulators can achieve low average error while violating identities that the modeled system satisfies exactly. Architectural constraint layers, output projection, probabilistic conditioning, and physics penalties address this problem in climate emulation, PDE surrogate modeling, and neural operators (Beucler et al., 2021; Hansen et al., 2023; Harder et al., 2023; Liu et al., 2024). Exact enforcement has an unambiguous benefit: every returned state satisfies the enforced invariant. A different claim—that the constraint itself improved predictive accuracy or out-of-distribution (OOD) generalization—requires a controlled attribution design.

That control is easy to miss. Autoregressive simulators may predict an absolute next state $f_\theta(x_t)$ or an update $x_t + g_\theta(x_t)$. Exact conservation is naturally implemented by projecting the update into the invariant tangent space. Comparing this hard residual model only with an unconstrained absolute-output model changes two factors at once: output parameterization and enforcement. Residual updates are themselves a strong inductive bias in learned simulation (Pfaff et al., 2021). A favorable comparison therefore does not identify which factor deserves credit.

We separate the predictive effects of coordinates, training enforcement and inference enforcement. The trained design crosses absolute/residual outputs with free/hard training; toggling projection on each locked checkpoint completes an eight-cell cube without optimization. This also exposes that projected training identifies raw outputs only modulo the invariant direction: a checkpoint may be accurate with its projector and unstable without it despite one-step conserving equivalence. Thus

“plug-and-play” means safe same-checkpoint composition through rollout, not software compatibility or exact constraint satisfaction.

Our contributions are:

- an eight-cell intervention design that identifies separate coordinate, training and inference effects, including three-way dependence hidden by a bundled comparison;
- a projection-gauge non-identifiability result and a direct same-checkpoint intervention establishing material transfer from invariant-direction input changes to conserving dynamics;
- prospectively frozen 30-seed Advection and independently frozen 30-seed 2D shallow-water studies that establish material projector dependence in the tested FNO rollouts; the registered U-Net transfer and post-hoc deployment control define the observed architecture scope; and
- orthogonal error identities, a channel-separable training null and tangent-kernel bounds that distinguish mechanically removed drift from changes in learned conserving dynamics.

2 RELATED WORK

Encoding physical constraints. Hard constraints enforce analytic identities through architectures, invariant operators, projection, probabilistic correction, or learned correction (Beucler et al., 2021; Richter-Powell et al., 2022; Liu et al., 2023; 2024; Hansen et al., 2023; Mouli et al., 2024; Zhang et al., 2023; Harder et al., 2023; Tanaka et al., 2025; Liu et al., 2025). Most directly, Duruisseau et al. (2024) compare an unconstrained FNO, post-hoc projection, constraint finetuning, and projected training; a contemporaneous submission ablates tangent-space projection and residual guidance (Anonymous Authors, 2026). These establish end-to-end feasibility and accuracy, but do not remove projection from the same constrained checkpoint, cross output coordinates, and follow that same-weight intervention through autoregressive feedback. Our target is this training-by-deployment interaction, not a replacement enforcement architecture.

Gradient interaction in physics learning. Composite PINN objectives can produce severe gradient-scale imbalance or directional conflict; adaptive scaling, dual-cone updates, and second-order preconditioning have been proposed to address these optimization pathologies (Yao et al., 2023; Hwang & Lim, 2024; Wang et al., 2025). Our mechanism question is narrower and structurally different: the two squared-error channels are orthogonal in output space, but a shared neural operator can map them to non-orthogonal parameter gradients. We do not propose a new optimizer. We use their alignment and an exact free-versus-hard one-step intervention to test whether this parameter-space coupling helps explain attribution.

Neural PDE evaluation. Fourier neural operators (FNOs) learn mappings between function spaces and support evaluation at unseen discretizations (Li et al., 2021). PDEBench provides standardized time-dependent PDE data, baselines, and metrics, including conservation-aware diagnostics (Takamoto et al., 2022a). Our experiments use its public Advection and two-dimensional shallow-water data, but change the evaluation question: rather than ranking a new operator, we estimate paired effects of parameterization and enforcement under the same locked weights.

3 SEPARATING PARAMETERIZATION AND CONSERVATION EFFECTS

3.1 LINEAR INVARIANT AND ERROR CHANNELS

Let $x \in \mathbb{R}^n$ be the last observed state and y the true next state. For a nonzero $a \in \mathbb{R}^n$, the linear invariant is $a^\top y = a^\top x$. Define the orthogonal projectors

$$P = \frac{aa^\top}{\|a\|_2^2}, \quad Q = I - P. \quad (1)$$

For prediction $\hat{y}$ and error $e = \hat{y} - y$, Pe is the invariant-violating channel and Qe is the conserving channel. We report conserving RMSE, R_{cons} , from Qe . This metric deliberately does not reward a

method merely for removing the mean error; total squared error decomposes as $\|e\|_2^2 = \|Qe\|_2^2 + \|Pe\|_2^2$.

The least-squares projection of an absolute prediction onto the affine invariant set is

$$\Pi_x(\hat{y}) = \hat{y} - P(\hat{y} - x). \quad (2)$$

Proposition 1 (Projection decoupling). *If $a^\top y = a^\top x$, then $\Pi_x(\hat{y}) - y = Q(\hat{y} - y)$. Consequently, projection removes the violating channel exactly and leaves the conserving-channel error unchanged.*

The proof is in Appendix A. In the experiment, the post-hoc projection of the trained free model is therefore an implementation control, not an independent learned baseline.

Proposition 2 (Projected-training gauge). *For any raw predictor $z(x)$ and scalar function $\alpha(x)$, $\Pi_x(z(x) + \alpha(x)a) = \Pi_x(z(x))$. Hence, in any hypothesis class closed under this transformation, a loss applied only after projection cannot identify the raw predictor’s invariant-channel component. Removing the projector from a hard-trained checkpoint selects an unidentified representative of the same deployed predictor.*

The proof is in Appendix B. Proposition 1 still makes the projected and unprojected predictions agree in conserving error for the same one-step input. For our FNO, the bias in its final pointwise map realizes at least every constant shift αa , so the closure condition is not merely an unrestricted-function abstraction. Autoregressive equivalence does not follow: an unprojected invariant-channel error changes the next input, where the network’s conserving response can differ. This yields a falsifiable signature of gauge exposure followed by state feedback—one-step equality followed by multi-step divergence.

Exact satisfaction of one invariant need not imply physical admissibility under another constraint. This distinction matters for conserved nonnegative fields such as water depth. Write $m(v) = n^{-1}\mathbf{1}^\top v$ for the uniform-grid mean.

Proposition 3 (Conservation–positivity compatibility). *For the uniform-mass invariant, $\Pi_x(z) = z + [m(x) - m(z)]\mathbf{1}$. If $z \geq 0$, the projected state is componentwise nonnegative if and only if $\min_i z_i \geq m(z) - m(x)$. Hence an orthogonal mass projection can enforce conservation exactly while introducing negative state values.*

This exact criterion motivates reporting invariant drift, conserving-channel forecast error, and admissible-set violations as three separate outcomes. In particular, clamping a projected field would silently compose two interventions and destroy the clean training-by-inference factorial.

The fixed-prediction statement also supplies a training-time null model. Let a coordinate choice $s \in \{A, R\}$ have base map $b_A(x) = 0$ or $b_R(x) = x$, and write the learned raw map as $z_\theta(x)$. Free and hard training have the same conserving prediction $Q(b_s(x) + z_\theta(x))$; they differ only because the free loss also contains the violating channel.

Proposition 4 (Channel-separable training null). *Suppose $\theta = (\theta_Q, \theta_P)$ and $z_\theta = z_Q(\theta_Q) + z_P(\theta_P)$, where $Pz_Q = 0$, $Qz_P = 0$, the regularizer separates over the two parameter blocks, and paired free and hard runs use identical minibatches, initial θ_Q , and a blockwise deterministic optimizer. Then their θ_Q iterates and conserving predictions are identical. Consequently, exact enforcement has zero conserving-channel training effect in that coordinate system.*

The proof is in Appendix D. A nonzero inference-matched training credit therefore rejects this separable-training null: training through the projector changed the admissible prediction through shared parameters, regularization, or optimization. A nonzero coordinate-by-training interaction makes the stronger statement that this coupling depends on absolute versus residual output coordinates. A bundled hard-versus-free rollout contrast is not enough for either conclusion because per-step projection also changes later inputs.

A local gradient diagnostic complements the training null. Shared parameters can couple the orthogonal output channels through the off-diagonal tangent-kernel block $\tilde{Q}K_\theta\tilde{P}$. Appendix E states the exact first-order transfer identity and a finite-step remainder bound. Because the actual optimizer is Adam, the empirical test also measures an exact matched one-Adam-step difference; neither diagnostic is assumed to explain the final rollout interaction.

3.2 MECHANISMS, PATH DEPENDENCE, AND PATH-AVERAGED CREDIT

The trained FNO body and parameter count are identical across mechanisms. With raw network output $g_\theta(x)$, the relevant arms are

$$\text{free} : \hat{y} = f_\theta(x), \quad (3)$$

$$\text{free_res} : \hat{y} = x + g_\theta(x), \quad (4)$$

$$\text{hard_abs} : \hat{y} = \Pi_x(f_\theta(x)), \quad (5)$$

$$\text{hard} : \hat{y} = x + Qg_\theta(x), \quad (6)$$

$$\text{projection} : \hat{y} = \Pi_x(f_\theta(x)). \quad (7)$$

For `hard_abs`, gradients pass through Π_x throughout training. For `projection`, θ is the already trained `free` parameter vector and projection is applied only as a derived evaluation transform; there is no fifth training run. The two confirmatory contrasts use conserving RMSE averaged over the same 30 seeds:

$$\Delta_{\text{param}} = R_{\text{cons}}(\text{free}) - R_{\text{cons}}(\text{free_res}), \quad \Delta_{\text{hard}} = R_{\text{cons}}(\text{hard}) - R_{\text{cons}}(\text{free_res}). \quad (8)$$

Positive Δ_{param} favors residual parameterization; negative Δ_{hard} favors hard enforcement. Together they give a path-specific decomposition of the bundled comparison, $R_{\text{cons}}(\text{free}) - R_{\text{cons}}(\text{hard}) = \Delta_{\text{param}} - \Delta_{\text{hard}}$. They do not estimate a fully crossed parameterization-by-training interaction.

Proposition 5 (Three-cell non-identification). *Without an additivity or other structural restriction on the missing potential outcome, observing only Y_{A0} , Y_{R0} , and Y_{R1} does not identify the factorial interaction or a unique path-averaged allocation. Infinitely many nonnegative values of the unobserved trained cell Y_{A1} preserve those three observations while giving distinct values of I_{path} and the two Shapley credits.*

The proof is in Appendix H. Consequently, neither a favorable three-arm sequence nor post-hoc projection of $A0$ can identify what training in the missing absolute-hard cell would have done. The projection is essential as an algebraic control, but it is not that trained intervention.

To identify that interaction, let $Y_{A0}, Y_{R0}, Y_{A1}, Y_{R1}$ denote paired-seed conserving RMSE for absolute versus residual coordinates (A, R) crossed with free versus hard enforcement $(0, 1)$. Define positive credits as error reductions,

$$P_0 = Y_{A0} - Y_{R0}, \quad P_1 = Y_{A1} - Y_{R1}, \quad (9)$$

$$E_A = Y_{A0} - Y_{A1}, \quad E_R = Y_{R0} - Y_{R1}. \quad (10)$$

The parameterization credit differs across enforcement levels by

$$I_{\text{path}} = P_0 - P_1 = E_A - E_R = Y_{A0} - Y_{R0} - Y_{A1} + Y_{R1}. \quad (11)$$

Thus I_{path} is simultaneously the factorial interaction and the discrepancy between two sequential credit-allocation paths. When it is not practically negligible, neither single path defines a unique contribution. We then use the exact two-factor Shapley averages

$$\phi_{\text{param}} = \frac{1}{2}(P_0 + P_1), \quad \phi_{\text{enforce}} = \frac{1}{2}(E_A + E_R), \quad \phi_{\text{param}} + \phi_{\text{enforce}} = Y_{A0} - Y_{R1}. \quad (12)$$

These are intervention credits computed from trained factorial cells, not feature-attribution scores. They average over the two possible orders instead of choosing the more favorable path after observing results (Shapley, 1953).

For the factorial primary cell, the smallest effect size of interest (SESOI) is $\delta = 0.1 \bar{Y}_{R0}$. The ordered frozen classification is: *material non-additivity* if the 95% paired-bootstrap interval for I_{path} lies wholly above $+\delta$ or below $-\delta$; *smaller statistical non-additivity* if that interval excludes zero but the first rule does not pass; *practical additivity* if the 90% interval lies wholly inside $[-\delta, +\delta]$; and *unresolved* otherwise. Thus both interaction and additivity are falsifiable, while an interval spanning both zero and a SESOI boundary receives no positive label. Seeds, not trajectories or rollout frames, are the inference units; each bootstrap draw resamples the complete paired four-cell seed vector. We use 50,000 draws. Twelve secondary interaction sign-flip tests receive Holm correction but cannot override the primary classification.

3.3 SEPARATING TRAINING ENFORCEMENT FROM INFERENCE PROJECTION

For multi-step rollouts, let Y_{cte} denote conserving RMSE in coordinate $c \in \{A, R\}$, with projector use during training $t \in \{0, 1\}$ and at every inference step $e \in \{0, 1\}$. The four trained models give $Y_{A00}, Y_{R00}, Y_{A11}, Y_{R11}$. Changing only the compatible forward map at evaluation supplies the other four cells; all such records take zero optimizer steps. Define the coordinate contrast $D_{te} = Y_{Ate} - Y_{Rte}$. Then

$$I_{\text{bundle}} = D_{00} - D_{11}, \quad (13)$$

$$T_0 = D_{00} - D_{10}, \quad T_1 = D_{01} - D_{11}, \quad (14)$$

$$E_0 = D_{00} - D_{01}, \quad E_1 = D_{10} - D_{11}, \quad (15)$$

$$J = T_0 - T_1 = E_0 - E_1 = D_{00} - D_{10} - D_{01} + D_{11}. \quad (16)$$

Here J is the coordinate-by-training-by-inference interaction. The path-averaged credits

$$\phi_{\text{train}} = \frac{1}{2}(T_0 + T_1), \quad \phi_{\text{infer}} = \frac{1}{2}(E_0 + E_1) \quad (17)$$

obey $\phi_{\text{train}} + \phi_{\text{infer}} = I_{\text{bundle}}$. We froze this cube while the fresh factorial was incomplete and no metric had been accessed. Its primary cell, bootstrap, SESOI, and ordered classification are inherited from the two-factor experiment. The local gradient diagnostic is matched to T_1 , where both endpoints use projected inference; its association with I_{bundle} is retained only as a labeled secondary analysis.

4 PDEBENCH PROTOCOLS

Data and invariant. We use PDEBench 1D Advection with velocity $\beta = 0.4$ and periodic boundary conditions (Takamoto et al., 2022a;b). The native tensor contains 10,000 trajectories, 201 used frames, and 1024 spatial cells. The spatial mean (equivalently, uniform-grid mass) is invariant. A frozen admission gate checks file hashes, schema, finiteness, and target-mean drift. To construct 512- and 256-cell fields without changing the invariant, we average consecutive blocks of two and four native cells in float64 before casting to float32; this restriction preserves the discrete mean.

Trajectories 0–255 are the external test set, 1000–8999 form the training pool, and 9000–9255 were used only for outcome-blind data preflight. Each seed samples 2,048 unique training trajectories. Inputs are 10 frames and the target is the next frame at temporal stride five (physical time 0.05). Models train only on 256 cells and are evaluated without retraining on ID 256 and OOD 512 and 1024 cells at autoregressive horizons $\{1, 4, 16, 31\}$.

Model and optimization. The FNO has width 20, 12 Fourier modes, four spectral-plus-pointwise blocks, padding two, GELU activations, and projection width 128 (Li et al., 2021). Every trained arm uses Adam with learning rate 10^{-3} , weight decay 10^{-4} , batch size 50, 200 epochs, and a StepLR decay of 0.5 at epoch 100. There is no validation selection, early stopping, or hyperparameter search. Arms within a seed share the sampled trajectories, minibatch order, training windows, and initial parameter tensor. The formal factorial runs all four trained cells from identical initial tensors with fresh seeds 3000–3029; no earlier checkpoint, seed, or record is reused. Its protocol and analysis rule were frozen after a path-specific parent study but before seed 2999 preflight or any formal model run. The estimates below use only the declared factorial and transfer cohorts.

After observing the FNO result but before any U-Net record existed, we froze a transfer test that changes only the model body. The two-level fully convolutional 1D U-Net (Ronneberger et al., 2015) uses base width 20, GroupNorm, SiLU, average-pooling, transposed-convolution upsampling, skip connections, and spatial coordinates. Data, splits, training arms, optimization, evaluation cells, and decision rule are unchanged, with fresh paired seeds 7000–7029. Its fixed gate requires a material primary J and at least 6/12 material cells. This is neither a capacity-matched architecture effect (FNO: 23,937 parameters; U-Net: 66,941) nor a claim that a finite-receptive-field U-Net is a discretization-invariant operator. The post-hoc forecast-skill analysis below separately tests whether these checkpoints improve on persistence.

Before inspecting any factorial metric, we separately froze a mechanism test that reconstructs each seed’s exact epoch-zero first minibatch. In absolute and residual coordinates it decomposes the full loss into $L_Q + L_P$, records $\langle g_Q, g_P \rangle$ and its cosine, and executes matched free and hard one-step Adam updates from the identical initial tensor. For each coordinate system, we compute the post-step

free-minus-hard conserving-loss difference. The local interaction subtracts the residual-coordinate difference from the absolute-coordinate difference. The frozen primary test is the paired-seed Spearman association between that local interaction and the inference-matched training interaction T_1 , with a 50,000-draw 95% bootstrap interval. The original association with the bundled interaction remains secondary. A non-positive or zero-crossing interval does not support the proposed local mechanism.

After the cube was complete and that gradient association was unsupported, we froze a targeted same-checkpoint gauge intervention before any new record existed. From one ground-truth history it computes a hard checkpoint’s compatible raw state z_1 and projection p_1 , appends each to an otherwise identical history, and compares the conserving channel of the next raw prediction. The primary per-seed estimand is this feedback RMSE divided by the projected-history branch’s conserving RMSE, averaged across coordinates at OOD 512; materiality requires its 95% interval to lie wholly above 0.10. The exact 60-record run crosses 30 seeds and two coordinates with zero optimization. The numerical identity criterion uses a 10^{-6} absolute floor together with a worst-case IEEE-754 bound; non-invariant roundoff must remain below 0.1% of the projected baseline.

Prospective primary and integrity gates. The primary cell was fixed as OOD 512 at horizon 16; the other 11 cells were mandatory secondary results. The core contract is exactly 150 records: 30 seeds times four trained arms plus one zero-training projection. After its one-shot analyzer passed, all 120 trained checkpoints were hash-locked and used to derive the three missing cube cells per seed, exactly 90 additional records with zero optimization. The local gradient diagnostic and direct gauge intervention each add exactly 60 coordinate-by-seed records. Before each analyzer may run, it verifies record coverage, run-ID uniqueness, source and configuration hashes, seed pairing, parameter parity, finite metrics, checkpoint binding, the target invariant, and the horizon-one projection identity. The original confirmation intervals use the seed as inference unit; secondary sign-flip tests receive Holm correction across the 12 mandatory cells.

Independent shallow-water replication. Before downloading or inspecting its values, we separately froze PDEBench’s 2D radial-dam-break shallow-water task (version 8, 2D_rdb_NA_NA.h5). Its 1,000 trajectories contain 101 frames on a 128×128 non-periodic grid. Water depth supplies both a linear invariant (uniform-grid mass) and an independent admissibility outcome (nonnegativity). A single-field FNO2d with width 20 and eight modes per axis takes ten depth fields plus coordinates and predicts the next field; no normalization or clamp is applied. Each seed samples 512 trajectories from groups 200–999, trains at 64×64 for 100 epochs, and is evaluated on held-out groups 0–127 at ID-64 and native OOD-128, horizons $\{1, 4, 16, 31\}$. Seeds 6000–6029 use the same four trained arms and inference-toggle cube, giving exactly 150 core records, 90 zero-training derived records, and 120 locked checkpoints. OOD-128/horizon-16 is primary. OOD-128/horizon-31 prospectively tests whether projection’s mass gain trades against or improves negative-depth deficit, while reporting conserving RMSE as a third, dynamics-specific outcome.

5 RESULTS

5.1 THE FOUR TRAINED CELLS SHOW A SMALL BUNDLED INTERACTION

In the fresh four-cell factorial, the primary OOD-512, horizon-16 cell shows that residual coordinates reduce conserving RMSE by 0.00154 [0.00080, 0.00231] under free training and by 0.00343 [0.00195, 0.00531] under hard training. Their difference, $I_{\text{bundle}} = -0.00189 [-0.00373, -0.00025]$, excludes zero but crosses the frozen SESOI boundary -0.00322 . Its ordered classification is therefore *smaller statistical non-additivity*, not material non-additivity. The two-factor Shapley credits are both positive: $\phi_{\text{param}} = 0.00248 [0.00163, 0.00357]$ and $\phi_{\text{enforce}} = 0.00096 [0.00023, 0.00167]$. On these four deployed cells alone, both modeling choices appear useful and only mildly path dependent.

5.2 THE COMPLETE CUBE REVEALS PROJECTOR DEPENDENCE HIDDEN BY CANCELLATION

The missing inference cells reverse that interpretation (Figure 1). Applying the projector to free-trained checkpoints barely moves the primary means: $Y_{A00} = 0.03373$ versus $Y_{A01} = 0.03375$,

Table 1: Fresh-seed primary effects. Positive P_0 and P_1 favor residual coordinates. I_{bundle} is the four-trained-cell interaction; J is the full coordinate-by-training-by-inference interaction. Intervals are 95% paired bootstrap CIs. The bundled interval follows the factorial primary analysis.

Estimand	Mean	95% CI	Frozen interpretation
P_0	0.00154	[0.00080, 0.00231]	residual favored
P_1	0.00343	[0.00195, 0.00531]	residual favored
I_{bundle}	-0.00189	[-0.00373, -0.00025]	smaller statistical non-additivity
J	0.22974	[0.19644, 0.26345]	material non-additivity
ϕ_{train}	0.11302	[0.09634, 0.12975]	countervailing credit
ϕ_{infer}	-0.11491	[-0.13166, -0.09818]	countervailing credit

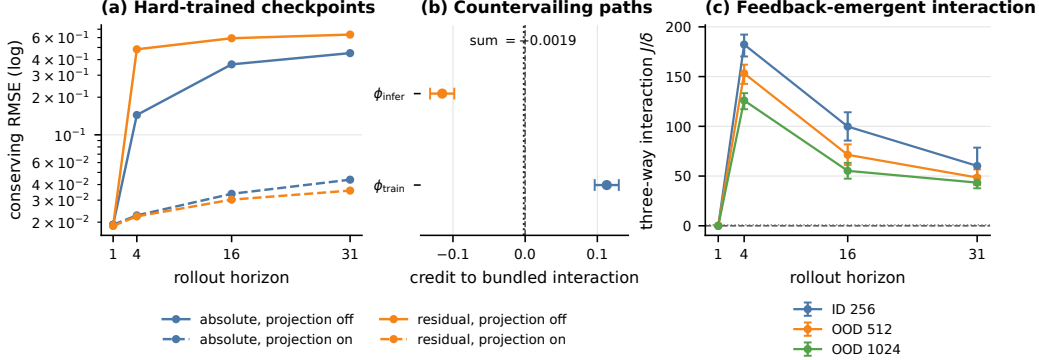


Figure 1: The projector becomes part of the learned autoregressive system. (a) At OOD 512, projected and unprojected hard-checkpoint evaluations agree in conserving RMSE at horizon one, as Proposition 1 requires, then diverge after feedback. Solid lines turn projection off; dashed lines retain it. (b) The primary path-averaged training and inference credits are large, opposite-signed, and sum to the small bundled interaction (vertical dotted line). (c) The three-way interaction, normalized by each cell’s SESOI, is near zero at horizon one and material from horizon four at every resolution. Error bars are 95% paired CIs.

and $Y_{R00} = 0.03219$ versus $Y_{R01} = 0.03218$. Removing it from hard-trained checkpoints instead raises conserving RMSE from $Y_{A11} = 0.03372$ to $Y_{A10} = 0.36627$, and from $Y_{R11} = 0.03029$ to $Y_{R10} = 0.59262$. These are evaluations of the same locked weights through compatible forward maps, with zero optimization.

Consequently, $J = 0.22974$ [0.19644, 0.26345] is 71.4 times the frozen SESOI and passes the material-nonadditivity rule; its Holm-adjusted sign-flip p -value is 0.00012. All 9 resolution cells at horizons 4, 16, and 31 are materially non-additive, whereas none of the three horizon-one cells is. The path-averaged contributions $\phi_{\text{train}} = 0.11302$ and $\phi_{\text{infer}} = -0.11491$ are individually large and opposite-signed, leaving only $I_{\text{bundle}} = -0.00189$ after cancellation. Thus the apparently benign deployed comparison does not imply modularity. Proposition 2 explains why hard training need not determine a usable raw invariant channel; the horizon-one identity followed by divergence in every registered horizon-four cell localizes the observed damage to autoregressive state feedback after that gauge is exposed.

5.3 U-NET RESULTS DEPEND ON THE DEPLOYMENT STRATEGY

The registered native-grid U-Net transfer has primary $J = -0.08415$ [-0.26665, 0.13334], classified *unresolved*, with only 3/12 material cells versus the 6/12 requirement. Its gate therefore *did not pass*. Post-hoc replay further places all eight OOD-512/horizon-16 cells below persistence, while deployed residual FNO skill is 0.9982 [0.9979, 0.9986] (Table 8). This native-grid recipe does not compare two predictively successful architectures; the registered results remain unchanged.

A separate post-hoc deployment control uses the same 120 U-Net checkpoints. At every step it block-averages history onto 256 cells, evaluates the U-Net there, and interpolates the output back to the target cell centers using periodic boundary conditions before feedback. No weights change. On the same 256 observed evaluation trajectories, the four trained/deployed arms have OOD-512/horizon-16 skill 0.942–0.986, and $J = 0.351$ [0.198, 0.556] under paired-seed resampling. The crossed seed/trajectory interval is [0.197, 0.555]. Thus a predictively useful non-FNO procedure can also exhibit path dependence. This is a comparison of deployment procedures, selected using separate development trajectories for predictive competence; it is not independent architecture confirmation or a pure model-body effect. Appendix P reports all cells, including the raw hard-checkpoint policies, both interval definitions and paired changes.

5.4 INDEPENDENT SHALLOW WATER REPRODUCES PATH DEPENDENCE

The prospectively frozen 2D shallow-water replication passes the same admission rule on a different equation, dimension, boundary geometry, and linear mass invariant. Its ordinary four-trained-cell interaction is 0.02954 [0.01534, 0.04369]. Completing the cube instead gives $J = -86.69343$ [-107.65055, -66.68345], whose magnitude is 5777.2 times the frozen SESOI (Holm $p = 8.000e-05$). The interaction is practically additive at horizon one at both resolutions, then material at horizons 4, 16, and 31, for 6/8 mandatory cells.

The diagnostic failure is sharply coordinate dependent: removing the projector from the hard-trained absolute checkpoint raises mean conserving RMSE from 0.09507 to 86.79537, whereas the corresponding residual checkpoint is unchanged at the reported precision (0.15006 in both conditions). The path-averaged training and inference credits are again large and opposing, -43.32405 and 43.35358, summing to the much smaller bundled interaction. This is not a claim that the unprojected hard model is a sensible deployment mode; it is evidence that the projector is constitutive of what projected training learned. Post-hoc paired-seed summaries show median absolute-coordinate removal harm of 74.71 (IQR 42.71–128.71), positive in 28/30 seeds within this fixed test cohort (Appendix M).

The independent admissibility test finds joint improvements. At OOD-128/horizon-31, adding projection to free-trained checkpoints improves maximum mass drift by 0.46656 [0.35540, 0.58242], conserving RMSE by 0.06994 [0.04168, 0.10034], and mean negative-depth deficit by $9.864e-05$ [3.640e-08, 0.00028]. Under the frozen sign convention and rule, this is a *resolved conservation-positivity synergy*, not a conservation-positivity trade-off.

5.5 A DIRECT INTERVENTION CONFIRMS GAUGE-TO-DYNAMICS FEEDBACK

The two-step same-checkpoint intervention isolates the proposed feedback link rather than inferring it from rollout timing. The first histories differ only by the raw prediction’s uniform invariant component; both then pass through the same compatible raw forward map. In the primary OOD-512 cell, the conserving-channel response has feedback RMSE 0.17447 [0.15881, 0.19064] and a feedback-to-projected-baseline ratio of 8.68398 [7.91007, 9.50538]. The latter is 86.8 times the frozen 0.10 practical threshold and is classified as *material gauge feedback*. All 3/3 registered resolutions reach the same classification, and both absolute and residual coordinates do so separately. The maximum numerical contamination is only $6.549e-05$ of the projected baseline, below the prespecified 0.00100 gate.

The intervention establishes material transfer from the invariant input direction into the learned conserving dynamics. Across seeds, its Spearman association with coordinate-averaged horizon-16 removal harm is $\rho = 0.28765$ [-0.09679, 0.62675], classified *not supported*. The direct response identifies the feedback channel; the seed association leaves its two-step magnitude unvalidated as a predictor of later harm.

Physically consistent histories distinguish a second question. The gauge experiment compares two histories induced by projector removal. A post-hoc Advection analysis shifts all ten input frames by a common constant, or adds an analytically advecting sinusoid, and compares the next-step response with the corresponding physical response. For whole-history constant shifts, the OOD-512 normalized conserving residual is 0.0083–0.0106 across the four training arms. Hard-minus-free differences are positive in both coordinates: the hard-trained checkpoints have larger constant-shift

equivariance errors on these histories (Table 11). This descriptive comparison reuses the checkpoints and tests a physically consistent history perturbation separately from projector removal. Appendix O gives amplitudes, analytic references and uncertainty.

Initial gradient interaction and final training effects. The frozen first-minibatch intervention finds an absolute-minus-residual one-Adam-step effect of 0.00015 [0.00012, 0.00019], with the interval excluding zero. Its paired-seed association with the inference-matched final training interaction $T_1 = -0.00186$ is $\rho = 0.062$ with 95% bootstrap CI $[-0.297, 0.398]$, classified *not supported*. The analogous bundled association is also unsupported. The intervention detects a local training interaction; its seed-level association with the final effect remains uncertain. The material feedback result comes from the direct input intervention.

Conserving error isolates learned predictive changes. Hard and projected rollouts enforce the invariant. Conserving RMSE measures the orthogonal channel that fixed projection leaves unchanged (Proposition 1). Changes in this error therefore identify effects of training and autoregressive feedback beyond the mechanically removed invariant drift.

6 IMPLICATIONS AND CONCLUSION

Projection participates in the learned transition. The two frozen FNO studies establish material dependence on the inference projector. In Advection, same-checkpoint projection leaves conserving error unchanged at horizon one, while removing it produces material interactions at every registered horizon-four cell. The direct input intervention confirms that the exposed invariant component changes the next conserving prediction. Together, the intervention cube and the input test locate a learned feedback pathway through the projector.

Bundled comparisons conceal opposing effects. The Advection training and inference credits are large and opposite-signed, leaving a small bundled interaction after cancellation. The eight-cell design recovers these separate effects by adding and removing projection on both free- and hard-trained checkpoints with fixed weights. The projection-gauge result explains the corresponding identification limit: the projected loss leaves the raw invariant component unconstrained. Raw outputs require their own predictive validation when deployment bypasses the projector.

Evaluate conserving dynamics as well as exact enforcement. Conserving error $\|Qe\|$ measures predictive changes outside the mechanically corrected channel, while invariant error $\|Pe\|$ measures enforcement. Both are needed to assess rollout changes; positivity requires its separate compatibility condition (Proposition 3). The registered U-Net result and its post-hoc resolution adapter also make deployment strategy part of the empirical scope. In the two frozen FNO studies, conservation enforcement is a component of the trained dynamics. Evaluating it jointly during training and rollout exposes dependence that the ordinary bundled comparison misses.

7 LIMITATIONS

Evidence spans two public datasets, one coefficient/scenario per equation, uniform mass, and resolution shifts plus post-hoc Advection history interventions; it does not cover nonlinear invariants, generic noise, coefficient shifts, or broad initial-condition distributions. Both independent positive PDE studies use FNOs. The larger native-grid U-Net does not pass its frozen gate and is below persistence in the primary cell. Its predictive post-hoc resolution adapter changes deployment as well as effective spatial processing; neither comparison is capacity matched, and independent architecture scope remains open. The shallow-water surrogate observes depth but not momentum or velocity, making its evidence partial-state rather than full-system fidelity. Projector removal is diagnostic, and gauge non-identification does not force every optimizer to select an unstable representative. Credits, the training null, and the 10% SESOI condition on the fixed design; cells share trained models, and intervals condition on the frozen trajectories. The post-hoc crossed bootstrap additionally resamples those observed trajectories; it does not establish coverage for every initial-condition distribution. Constant-shift and advecting-sine symmetries apply here to linear periodic Advection, not to shallow-water depth.

REPRODUCIBILITY STATEMENT

Appendix J records the data version, split, conservative restriction, model, optimization, seeds, inference units, and frozen decision rules. The supplementary artifact contains the Hydra configurations, immutable data-file and checkpoint locks, 450 core records, 270 zero-training cube records, 60 local-gradient records, 60 direct-gauge records, validation tests, one-shot analyzers, and the exact JSON used to generate every reported empirical estimate and Figure 1. Checkpoint-lock manifests bind all 360 final weight files by path, byte count, payload identity, and SHA-256; this does not imply that the weight bytes are embedded in the PDF supplement. All formal records bind to source and protocol hashes. The accompanying revision data view contains all 192 checkpoint-skill cells and 144 physical-history response cells, both interval definitions, and generated tables. Their post-hoc status remains separate from the original confirmation. The resolution-adaptor view additionally contains 192 native/adapted skill cells and paired trajectory errors for CPU replay. Full checkpoints and trajectory arrays are retained in the recovery archive.

AI USE STATEMENT

Generative AI tools were used to help formulate and critically revise the attribution framework, mathematical statements and proof drafts, hypotheses and the prospective experiment protocol; implement and test experiment and analysis code; prepare the conservative data interface; interpret frozen outputs; discover and summarize related literature; generate figure code; format references; and draft and edit the manuscript. They were not used to generate the underlying PDEBench observations. The authors reviewed the primary sources, executable tests, immutable hashes, analysis outputs, mathematical claims, and all AI-assisted text and code. The authors take responsibility for the final content, including claims and artifacts produced with AI assistance.

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A PROOF OF PROJECTION DECOUPLING

Because the target satisfies the invariant, $P(y - x) = 0$. Therefore

$$\Pi_x(\hat{y}) - y = \hat{y} - P(\hat{y} - x) - y \quad (18)$$

$$= e - P((\hat{y} - y) + (y - x)) \quad (19)$$

$$= e - Pe = Qe. \quad (20)$$

Applying P gives zero and applying Q returns Qe . Orthogonality of P and Q gives the squared-error decomposition.

B PROOF OF THE PROJECTED-TRAINING GAUGE

Because $Pa = a$ for the rank-one projector onto $\text{span}(a)$,

$$\Pi_x(z + \alpha a) = z + \alpha a - P(z + \alpha a - x) \quad (21)$$

$$= z - P(z - x) + \alpha(a - Pa) \quad (22)$$

$$= \Pi_x(z). \quad (23)$$

Thus every realizable raw map in the equivalence class $\{z + \alpha a : \alpha : \mathcal{X} \rightarrow \mathbb{R}\}$ induces exactly the same projected prediction and loss. Projected observations cannot distinguish members of this class, so the raw invariant-channel representative is unidentified without an additional restriction. This is an output-space statement; shared parameters may still couple the identified and unidentified channels, as analyzed by Propositions 4 and 6.

C PROOF OF CONSERVATION-POSITIVITY COMPATIBILITY

For $a = \mathbf{1}$, the projector subtracts the prediction’s mean excess: $\Pi_x(z)_i = z_i + m(x) - m(z)$. Therefore $\Pi_x(z)_i \geq 0$ for every coordinate exactly when $z_i \geq m(z) - m(x)$ for every coordinate, which is equivalent to $\min_i z_i \geq m(z) - m(x)$. Taking any nonnegative z with $0 \leq \min_i z_i < m(z) - m(x)$ proves that exact mass projection can create a negative component.

D PROOF OF THE CHANNEL-SEPARABLE TRAINING NULL

For either coordinate choice, the conserving component of both the free and hard prediction is $Q(b_s + z_Q(\theta_Q))$ because $Qz_P = 0$. The free squared loss decomposes orthogonally into a term depending only on θ_Q and a violating-channel term depending only on θ_P ; the hard loss retains the first term and makes the second constant. The separable regularizer preserves this equality of the θ_Q objectives. Thus paired minibatches give identical gradients for θ_Q at every step. Equal initialization and a blockwise deterministic optimizer imply by induction that all θ_Q iterates, and hence all conserving predictions, are identical. The paired conserving-error difference is therefore zero.

E LOCAL CHANNEL-TRANSFER STATEMENTS

The first optimization step yields a sharper local diagnostic. Decompose the free squared loss as $L = L_Q + L_P$ and write $g_Q = \nabla L_Q(\theta)$ and $g_P = \nabla L_P(\theta)$.

For a stacked minibatch prediction h_θ , let $e = h_\theta - y$, let J_θ be its parameter Jacobian, and let $\bar{P}$ and $\bar{Q}$ apply P and Q independently to every example. The finite-width empirical tangent kernel is $K_\theta = J_\theta J_\theta^\top$ (Jacot et al., 2018). Take $L_Q = \frac{1}{2} \|\bar{Q}e\|^2$ and $L_P = \frac{1}{2} \|\bar{P}e\|^2$; a common normalization only rescales the step size.

Proposition 6 (Tangent-kernel channel transfer). *Let $h_\theta^{\text{lin}}(\theta + u) = h_\theta + J_\theta u$. From the same parameters, take a free gradient step $\theta_F^+ = \theta - \eta(g_Q + g_P)$ and a hard step $\theta_H^+ = \theta - \eta g_Q$. Then*

$$\bar{Q}[h_\theta^{\text{lin}}(\theta_F^+) - h_\theta^{\text{lin}}(\theta_H^+)] = -\eta \bar{Q} K_\theta \bar{P} e, \quad (24)$$

$$\langle g_Q, g_P \rangle = (\bar{Q}e)^\top K_\theta (\bar{P}e). \quad (25)$$

Thus the violating-channel gradient has zero first-order effect on the conserving prediction for every violating residual if and only if the off-diagonal channel block $\bar{Q}K_\theta\bar{P}$ vanishes.

The proof is in Appendix F. This identity links the global separable-training null to a measurable local mechanism. Shared parameters need not produce a nonzero off-diagonal block, but any nonzero block is a route by which hard enforcement can change admissible predictions even though P and Q are orthogonal in output space.

Proposition 7 (One-step gradient coupling). *If ∇L_Q is β -Lipschitz, then a gradient-descent step of size η obeys*

$$L_Q(\theta - \eta(g_Q + g_P)) - L_Q(\theta - \eta g_Q) = -\eta\langle g_Q, g_P \rangle + r, \quad (26)$$

where $|r| \leq \frac{\beta\eta^2}{2} (\|g_Q + g_P\|_2^2 + \|g_Q\|_2^2)$.

The proof is in Appendix G. Negative gradient alignment therefore makes removing the violating-channel gradient locally favorable for L_Q , up to the stated remainder; positive alignment reverses that direction. Output coordinates can change g_P , providing a testable local mechanism for the factorial interaction. Because the actual optimizer is Adam and training is nonlinear, we prospectively record both the gradient geometry and the exact matched one-Adam-step difference; neither is assumed to explain the final rollout interaction. This use of gradient alignment follows the broader composite-loss diagnostic literature (Hwang & Lim, 2024; Wang et al., 2025), while the orthogonal conservation channels and matched enforcement intervention are specific to our design.

F PROOF OF TANGENT-KERNEL CHANNEL TRANSFER

The two parameter displacements differ by $\theta_F^+ - \theta_H^+ = -\eta g_P$. Because $g_Q = J_\theta^\top \bar{Q}e$ and $g_P = J_\theta^\top \bar{P}e$, the linearized conserving-output difference is

$$\bar{Q}J_\theta(\theta_F^+ - \theta_H^+) = -\eta\bar{Q}J_\theta J_\theta^\top \bar{P}e = -\eta\bar{Q}K_\theta\bar{P}e. \quad (27)$$

The inner-product identity follows from $g_Q^\top g_P = (\bar{Q}e)^\top J_\theta J_\theta^\top (\bar{P}e)$. If $\bar{Q}K_\theta\bar{P} = 0$, the first expression vanishes for every violating residual. Conversely, vanishing for every vector in the range of $\bar{P}$ implies $\bar{Q}K_\theta\bar{P} = 0$.

G PROOF OF THE ONE-STEP GRADIENT-COUPPLING BOUND

For a function with β -Lipschitz gradient, the descent lemma gives $L_Q(\theta + u) = L_Q(\theta) + g_Q^\top u + r_u$ with $|r_u| \leq \beta\|u\|_2^2/2$. Apply it to $u = -\eta(g_Q + g_P)$ and $v = -\eta g_Q$, then subtract the two expansions. Their linear terms differ by $-\eta g_Q^\top g_P$ and the triangle inequality gives

$$|r_u - r_v| \leq \frac{\beta\eta^2}{2} (\|g_Q + g_P\|_2^2 + \|g_Q\|_2^2), \quad (28)$$

which is Equation 26.

H PROOF OF THREE-CELL NON-IDENTIFICATION

Fix arbitrary nonnegative observed values Y_{A0} , Y_{R0} , and Y_{R1} , and let $C = Y_{A0} - Y_{R0} + Y_{R1}$. For any $c \leq C$, choose

$$Y_{A1} = Y_{A0} - Y_{R0} + Y_{R1} - c. \quad (29)$$

This value is nonnegative, and substitution into Equation 11 gives $I_{\text{path}} = c$ while leaving every observed cell unchanged. The half-line $c \leq C$ contains infinitely many choices. Because both $P_1 = Y_{A1} - Y_{R1}$ and $E_A = Y_{A0} - Y_{A1}$ depend on this unobserved value, the Shapley credits in Equation 12 are also unidentified. Proposition 1 constrains the conserving error of a fixed projected prediction, not the parameters learned by training through the hard absolute-output layer, so projection cannot supply the missing intervention.

I COMPLETE FROZEN INTERVENTION-CUBE SUMMARY

Table 2 lists the three-way interaction in every mandatory resolution-horizon cell. It is not a set of twelve replications: all rows reuse the same paired checkpoints, and Holm correction is applied to their sign-flip tests. The frozen ordered classification uses each cell’s own $\delta = 0.1\bar{Y}_{R00}$.

Table 2: All mandatory PDEBench Advection intervention-cube cells. J/δ gives effect size relative to the prespecified practical threshold; p is Holm-adjusted.

Case	Horizon	J	95% CI	J/δ	Class	Holm p
ID 256	1	2.52×10^{-10}	$[4.29 \times 10^{-11}, 4.76 \times 10^{-10}]$	0.0	smaller	0.0932
ID 256	4	0.3256	[0.3041, 0.3434]	182.2	material	0.0001
ID 256	16	0.2205	[0.1893, 0.2522]	99.8	material	0.0001
ID 256	31	0.1667	[0.1162, 0.2175]	60.2	material	0.0001
OOD 512	1	1.10×10^{-10}	$[-1.96 \times 10^{-10}, 4.10 \times 10^{-10}]$	0.0	additive	0.7632
OOD 512	4	0.3397	[0.3164, 0.3593]	153.2	material	0.0001
OOD 512	16	0.2297	[0.1964, 0.2634]	71.4	material	0.0001
OOD 512	31	0.1914	[0.1612, 0.2237]	48.5	material	0.0001
OOD 1024	1	-1.24×10^{-10}	$[-3.89 \times 10^{-10}, 1.48 \times 10^{-10}]$	0.0	additive	0.7632
OOD 1024	4	0.3450	[0.3209, 0.3651]	126.0	material	0.0001
OOD 1024	16	0.2338	[0.2004, 0.2676]	55.3	material	0.0001
OOD 1024	31	0.2203	[0.1913, 0.2531]	43.4	material	0.0001

J PRIMARY EIGHT CELLS AND ARTIFACT IDENTITY

Table 3 makes the full primary intervention transparent. Training and inference projection are separate binary factors. The four off-diagonal inference cells reuse locked weights and take zero optimizer steps.

Table 3: Mean conserving RMSE in all eight OOD-512, horizon-16 cells across 30 paired seeds.

	Training	Inference	Absolute	Residual
free		projector off	0.03373	0.03219
free		projector on	0.03375	0.03218
hard		projector off	0.36627	0.59262
hard		projector on	0.03372	0.03029

For reference, the primary conserving-RMSE means are 0.03373 for `free`, 0.03219 for `free_res`, 0.03372 for `hard_abs`, 0.03029 for `hard`, and 0.03375 for free-trained projection. These descriptive quantities do not define extra hypothesis tests. A projection rollout feeds corrected states back into the FNO, so after its first step its inputs differ from an unprojected rollout; the fixed-input identity is checked at horizon one rather than assumed at horizon 16.

The dataset is DaRUS PDEBench version 8, file ID 255674, 1D_Advection_Sols_beta0.4.hdf5, released under CC BY 4.0. The observed file size is 8,232,966,952 bytes. Its content hashes are MD5: d595bbfd2c659df995a93cd40d6ea568 and SHA256: d973ff2bb3c2a5edf42957ff029b78672451532ebd6bf475848ff23cfd3ee3b6. The fresh core JSONL/analysis SHA-256 prefixes are d0c8257af992/15ab5898699e; the zero-training cube prefixes are bee99e1b15d0/d12dc6d088d2; and the local-diagnostic prefixes are fd5251a63f6d/e6edc0803477. The direct-gauge data/analysis prefixes are 285ba66aeb84/87ed9676386e. The supplementary manifest records and verifies the complete hashes rather than these display prefixes.

The original confirmation uses seeds as its inference unit; the post-hoc skill and history analyses also report crossed seed/trajectory intervals. All mechanisms within a seed use the same training subset, deterministic window cycle, minibatch order, initialization tensor, and evaluation trajectories. No model outcome was read before complete coverage and the integrity gates passed.

K REGISTERED U-NET ARCHITECTURE-TRANSFER BOUNDARY

Table 4 reports the complete mandatory transfer grid rather than only the three favorable ID rows. The gate required both a material OOD-512/horizon-16 primary and at least 6/12 material cells. The primary is *unresolved*, and only 3/12 cells are material, so the registered transfer *did not pass*. Six cells show smaller statistical non-additivity, one is practically additive, and two are unresolved. This pattern is a boundary result, not evidence of absence in every U-Net cell.

Table 4: All mandatory U-Net intervention-cube cells. J/δ uses the same prespecified practical threshold as the FNO experiment; p is Holm-adjusted.

Case	Horizon	J	95% CI	J/δ	Class	Holm p
ID 256	1	2.21×10^{-10}	$[8.98 \times 10^{-12}, 4.26 \times 10^{-10}]$	0.0	smaller	0.2483
ID 256	4	-0.0791	$[-0.1135, -0.0428]$	-27.3	material	0.0018
ID 256	16	0.4236	$[0.2522, 0.6406]$	26.5	material	0.0002
ID 256	31	0.8233	$[0.5034, 1.2144]$	23.1	material	0.0001
OOD 512	1	-5.99×10^{-10}	$[-1.15 \times 10^{-9}, -3.44 \times 10^{-11}]$	0.0	smaller	0.2483
OOD 512	4	-0.0469	$[-0.0619, -0.0322]$	-1.0	smaller	0.0001
OOD 512	16	-0.0841	$[-0.2666, 0.1333]$	-0.6	unresolved	1.0000
OOD 512	31	-0.0188	$[-0.3469, 0.3441]$	-0.1	unresolved	1.0000
OOD 1024	1	2.86×10^{-10}	$[-4.69 \times 10^{-10}, 1.08 \times 10^{-9}]$	0.0	additive	1.0000
OOD 1024	4	-0.0537	$[-0.0723, -0.0342]$	-0.8	smaller	0.0002
OOD 1024	16	-0.3506	$[-0.5808, -0.0727]$	-1.6	smaller	0.0747
OOD 1024	31	-0.6128	$[-1.1199, -0.0811]$	-1.7	smaller	0.1933

Table 5: Mean U-Net conserving RMSE in the eight primary OOD-512, horizon-16 cells across 30 paired seeds. Off-diagonal inference cells reuse locked weights and take zero optimizer steps.

Training	Inference	Absolute	Residual
free	projector off	0.86332	1.47514
free	projector on	0.86423	1.47043
hard	projector off	0.82239	1.45803
hard	projector on	0.83205	1.54622

The U-Net merged core/analysis SHA-256 prefixes are ae0dda1c95e5/6e3252b5a1d3; the 120-checkpoint lock is e7d43d3d3ce9; and the cube/analysis prefixes are 01a23a0b77ff/0e2ad4bdf6f9. The formal seeds are 7000–7029. Both fixed 15-seed workers exited before the core analyzer ran; the cube followed only after exact 150-record coverage and checkpoint-lock verification. The supplementary manifest records the full hashes.

L INDEPENDENT SHALLOW-WATER ARTIFACT IDENTITY

Table 6: All eight shallow-water primary OOD-128, horizon-16 cells. Entries are mean conserving RMSE across 30 paired seeds; off-diagonal cells change only the inference forward map of a locked checkpoint and take zero optimizer steps.

Training	Inference	Absolute	Residual
free	projector off	0.12461	0.15006
free	projector on	0.11774	0.15006
hard	projector off	86.79537	0.15006
hard	projector on	0.09507	0.15006

The replication uses DaRUS PDEBench version 8, file ID 133021, 2D_rdb_NA_NA.h5, with 6,626,098,972 bytes and hashes MD5:75d838c47aa410694bdc912ea7f22282 and SHA256:

28f0c33723d70eebb420fc170e94b675c18e032fb697dcef080e114ca9645e3a. The data gate verifies all 1,000 group names, tensor and coordinate schemas, finiteness, strict target positivity, mass drift, and both block-average restrictions. The merged core JSONL/analysis SHA-256 prefixes are 5e486167a2a8/d194df9d3f8d; the checkpoint lock prefix is 772c52f9dbaa; and the zero-training cube/analysis prefixes are 69dfa936fd83/f8492362b377. Exact hashes are retained in the supplementary manifest.

M POST-HOC PAIRED-SEED REMOVAL-HARM DISTRIBUTIONS

For each original checkpoint pair, removal harm is conserving RMSE without minus with the projector at the primary OOD resolution and horizon 16. Positive values indicate harm. These post-hoc summaries reuse the original records, preserve all 30 seeds, and condition on their common test trajectories. The bootstrap mean intervals below are descriptive and do not replace the original frozen interaction intervals or SESOI. Median, interquartile range and sign counts show the empirical seed distribution; they do not constitute another confirmatory test.

Table 7: Conserving-RMSE removal harm in each dataset’s field units. Magnitudes cannot be compared directly between Advection and shallow water. The complete mean/median/IQR/min/max summaries for all eight cells, both RMSE metrics, and all resolutions/horizons are in the accompanying scientific data view.

Cohort / coordinate	Mean [95% CI]	Median [IQR]	Positive seeds
Advection FNO / A	0.333 [0.299, 0.365]	0.314 [0.259, 0.433]	30/30
Advection FNO / R	0.562 [0.557, 0.569]	0.558 [0.553, 0.570]	30/30
Advection U-Net / A	-0.010 [-0.040, 0.021]	-0.010 [-0.067, 0.059]	13/30
Advection U-Net / R	-0.088 [-0.274, 0.137]	-0.274 [-0.429, 0.096]	8/30
Shallow-water FNO / A	86.700 [66.517, 107.565]	74.712 [42.707, 128.710]	28/30
Shallow-water FNO / R	2.11e-08 [1.80e-10, 4.28e-08]	0.000 [-5.33e-09, 1.27e-08]	9/30

N POST-HOC FORECAST SKILL ON THE LOCKED ADVECTION CHECKPOINTS

The 120 FNO and 120 U-Net checkpoints comprise four trained arms and 30 paired seeds per architecture. Two inference policies give all eight cells on the existing 256 shared trajectories. We compute aggregate total MSE across seeds, trajectories, and spatial cells before dividing by persistence MSE; this is not an average of per-trajectory skill ratios. Persistence repeats the last observed field, and the analytic control translates it at the known Advection velocity. This analysis reuses both weights and test trajectories and remains post-hoc.

The 10,000-draw crossed bootstrap independently resamples seeds and shared trajectories while preserving pairing across cells. Seed-only intervals condition on those trajectories; both sets are in the scientific JSON. Table 8 reports descriptive 95% crossed intervals for every primary cell. For the two architecture-specific R11 summaries, the separately adjusted intervals use 97.5% coverage; all other cell intervals are marginal, not simultaneous. A negative skill means that total forecast MSE exceeds persistence MSE. In particular, very negative A10 and R10 FNO skill measures raw hard-checkpoint removal damage, not deployed accuracy.

Table 8: Post-hoc total-MSE skill at OOD-512/horizon-16. The original conserving-RMSE cube and its registered gate remain unchanged. A/R denote absolute/residual outputs; the two digits indicate training and inference projection.

Cell	FNO skill [95% CI]	U-Net skill [95% CI]
A00	0.9973 [0.9968, 0.9978]	-0.4446 [-0.6119, -0.2976]
A01	0.9978 [0.9973, 0.9982]	-0.4309 [-0.5930, -0.2885]
A10	-280.06 [-479.01, -115.64]	-1.2798 [-1.6019, -1.0112]
A11	0.9978 [0.9973, 0.9982]	-0.3238 [-0.4650, -0.1976]
R00	0.9975 [0.9970, 0.9979]	-3.2884 [-4.0935, -2.6208]
R01	0.9980 [0.9976, 0.9984]	-3.2580 [-4.0574, -2.5961]
R10	-265.31 [-311.56, -228.54]	-58.45 [-79.68, -40.51]
R11	0.9982 [0.9979, 0.9986]	-3.7688 [-4.7886, -2.9304]

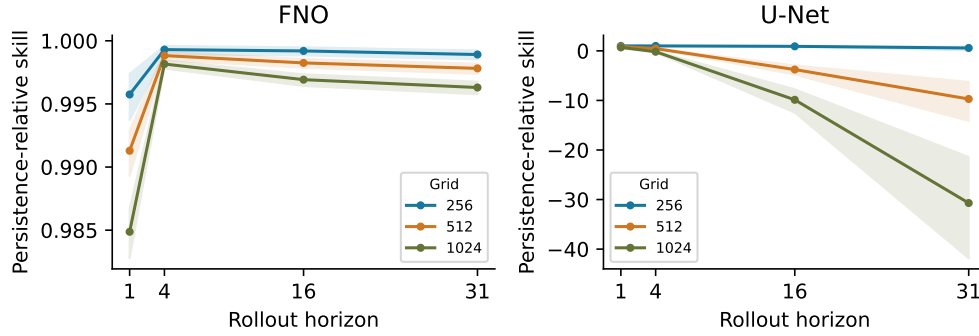


Figure 2: Deployed hard residual (R11) skill across all mandatory Advection grids and horizons. Shading gives pointwise 95% crossed intervals. Axes differ between architectures. This is post-hoc replay; the complete eight-cell estimates are in the accompanying CSV and JSON.

Table 9: All mandatory FNO cells for the four trained/deployed arms: total-MSE skill.

Grid/horizon	A00	A11	R00	R11
256/1	0.9930	0.9949	0.9938	0.9957
256/4	0.9991	0.9992	0.9992	0.9993
256/16	0.9985	0.9989	0.9988	0.9992
256/31	0.9972	0.9980	0.9980	0.9989
512/1	0.9906	0.9909	0.9913	0.9913
512/4	0.9987	0.9988	0.9988	0.9988
512/16	0.9973	0.9978	0.9975	0.9982
512/31	0.9952	0.9966	0.9959	0.9978
1024/1	0.9869	0.9852	0.9874	0.9849
1024/4	0.9981	0.9981	0.9981	0.9982
1024/16	0.9957	0.9964	0.9958	0.9969
1024/31	0.9926	0.9947	0.9934	0.9963

Table 10: Corresponding U-Net skill. This fixed recipe does not establish transfer among predictively useful architectures.

Grid/horizon	A00	A11	R00	R11
256/1	0.9954	0.9990	0.9993	0.9992
256/4	0.9948	0.9985	0.9974	0.9967
256/16	0.9540	0.9858	0.9340	0.9135
256/31	0.8735	0.9613	0.6895	0.5794
512/1	0.8055	0.8153	0.8669	0.8624
512/4	0.6545	0.6609	0.4826	0.4582
512/16	-0.4446	-0.3238	-3.2884	-3.7688
512/31	-0.1711	0.0482	-7.5920	-9.7065
1024/1	0.5747	0.5809	0.7058	0.6936
1024/4	0.3022	0.3074	-0.1280	-0.1759
1024/16	-0.3969	-0.2059	-8.4407	-9.8754
1024/31	-0.4227	0.0019	-26.39	-30.70

O POST-HOC PHYSICAL-HISTORY INTERVENTIONS

For linear periodic Advection, a constant shift of every history frame is a physical symmetry: the next field shifts by the same constant, so its conserving (Q) response is zero. For a sinusoid, the reference instead transports that perturbation at velocity $\beta = 0.4$ through both the ten history times and the next prediction time. Its analytic Q response is nonzero. For each existing FNO checkpoint we compare the deployed next-step responses with and without these interventions. We divide the spatial RMS of the response minus its analytic Q reference by $|\alpha|$, using amplitudes $\alpha/\sigma \in \{-0.5, -0.1, 0.1, 0.5\}$. The scale $\sigma = 0.7960596902$ is the standard deviation of native-grid values in the first ten frames of seed 3000’s 2,048 training trajectories; test values do not determine it.

All 120 checkpoints use the same 256 test trajectories and three resolutions. The primary post-hoc summary averages the four constant-shift residuals within seed at OOD-512, then averages seeds. The 98.75% crossed intervals adjust the four arm summaries. Paired hard-minus-free contrasts have descriptive 95% intervals and do not replace the frozen gauge test.

Table 11: Normalized whole-history constant-shift residual at OOD-512. Smaller values mean closer constant-shift equivariance, independently of exact output conservation.

Training	Normalized residual [98.75% CI]
Free absolute	0.0089 [0.0083, 0.0094]
Free residual	0.0083 [0.0081, 0.0086]
Hard absolute	0.0106 [0.0102, 0.0111]
Hard residual	0.0092 [0.0090, 0.0096]
Hard minus free	Difference [95% paired CI]
Absolute	0.0018 [0.0014, 0.0022]
Residual	0.0009 [0.0007, 0.0011]

Table 12: Advecting-sine residual at OOD-512, relative to the nonzero analytic conserving response. Intervals are descriptive 95% crossed bootstrap intervals.

Training	α/σ	Physical residual [95% CI]
Free absolute	-0.5	0.0144 [0.0136, 0.0153]
Free absolute	-0.2	0.0142 [0.0134, 0.0150]
Free absolute	+0.2	0.0142 [0.0134, 0.0152]
Free absolute	+0.5	0.0146 [0.0136, 0.0156]
Free residual	-0.5	0.0123 [0.0117, 0.0129]
Free residual	-0.2	0.0123 [0.0118, 0.0129]
Free residual	+0.2	0.0123 [0.0118, 0.0129]
Free residual	+0.5	0.0123 [0.0117, 0.0129]
Hard absolute	-0.5	0.0191 [0.0182, 0.0202]
Hard absolute	-0.2	0.0193 [0.0183, 0.0204]
Hard absolute	+0.2	0.0193 [0.0183, 0.0204]
Hard absolute	+0.5	0.0192 [0.0182, 0.0203]
Hard residual	-0.5	0.0138 [0.0131, 0.0145]
Hard residual	-0.2	0.0138 [0.0131, 0.0145]
Hard residual	+0.2	0.0138 [0.0131, 0.0145]
Hard residual	+0.5	0.0138 [0.0131, 0.0145]

A separate condition shifts only the last input frame. It is an input-sensitivity measurement, with no physically consistent history reference. At equal amplitude the ten-frame intervention has ten times its input energy, so their magnitude difference is not an equal-energy causal contrast. All signed amplitudes, resolutions, per-seed responses and marginal intervals are provided in the accompanying scientific JSON. Neither Advection symmetry is asserted for shallow-water depth.

P POST-HOC U-NET RESOLUTION-ADAPTER COMPARISON

This analysis changes deployment while preserving all 120 trained checkpoints. The wrapper restricts each input history to the training grid of 256 cells by block averaging. It evaluates the original network on that grid and interpolates its output to the target cell centers using periodic boundary conditions. The interpolated prediction enters the next history at the target resolution. For an integer refinement factor r , interpolation uses coordinate $(j + 1/2)/r - 1/2$ in coarse-cell indices and the adjacent periodic centers. Restriction and interpolation preserve the spatial mean. The wrapper is the identity operation at resolution 256; the paired replay arrays agree exactly there. It adds no fitted weights, but defines a different deployed autoregressive map.

One fixed wrapper was specified before evaluation replay. Development trajectories 9256–9511, disjoint from the training pool and original evaluation set, checked the four trained/deployed arms for seeds 7015 and 7016. All eight lower trajectory-bootstrap 95% skill bounds exceeded the fixed 0.5 competence threshold. This selected a deployment procedure, not a desired interaction. The subsequent comparison retains all 30 seeds, 256 previously observed evaluation trajectories, eight policies, three resolutions and four horizons. Reusing observed trajectories and checkpoints makes the analysis post-hoc; it does not change the original registered U-Net gate.

Table 13: Primary grid 512/horizon 16: total-MSE forecast skill. Adapted intervals jointly resample seeds and common trajectories. The raw A10/R10 policies remain diagnostic projector-removal interventions rather than trained standalone forecasts.

Cell	Native skill	Adapted skill	Adapted 95% CI
A00	-0.4446	0.9551	[0.9307, 0.9721]
A01	-0.4309	0.9685	[0.9547, 0.9791]
A10	-1.2798	-0.1295	[-0.4305, 0.1188]
A11	-0.3238	0.9862	[0.9819, 0.9897]
R00	-3.2884	0.9472	[0.9226, 0.9674]
R01	-3.2580	0.9478	[0.9230, 0.9682]
R10	-58.4482	-54.7274	[-76.1867, -36.7922]
R11	-3.7688	0.9422	[0.9124, 0.9645]

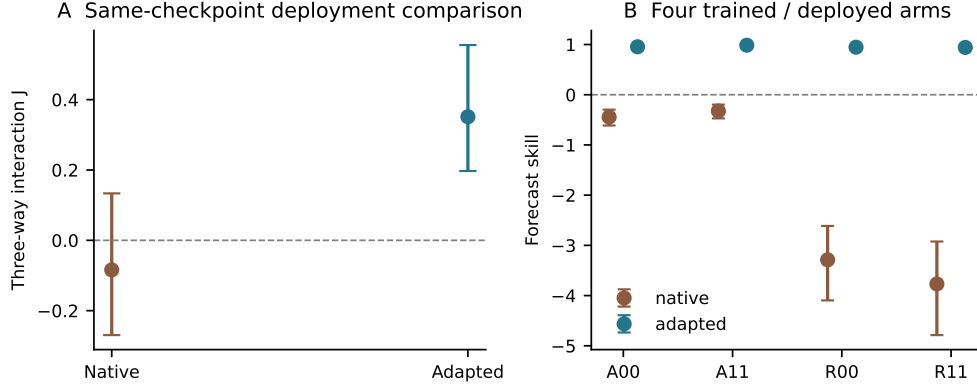


Figure 3: Post-hoc paired deployment comparison at grid 512/horizon 16. Bars are marginal crossed seed/trajectory 95% intervals. The four trained/deployed arms improve on persistence with the adapter; raw projector-removal policies appear in the complete table.

For every bootstrap draw, all policies and both deployment strategies share sampled seeds and trajectories. Skill is a ratio of aggregate MSEs. The interaction instead uses each seed’s conserving RMSE and then averages the seed-level contrast, matching the original cube estimand. At the primary cell, adapted-minus-native J is 0.435 [0.262, 0.613] under crossed resampling. Paired skill changes and both interval definitions for every cell are in the scientific JSON. The table uses the descriptive threshold $\delta = 0.1\bar{R}_{00}$ within each deployment strategy; 9/12 adapted and 3/12 native cells meet this criterion. These are not additional registered gate decisions or multiplicity-adjusted confirmation tests.

Table 14: All resolution–horizon interactions. Marginal paired-seed intervals condition on the observed trajectories. Thresholds are strategy-specific; the original native-grid confirmation and its decision rule are retained separately.

Grid/horizon	Strategy	J	Seed 95% CI	δ	Material
256/1	native	0.0000	[-0.0000, 0.0000]	0.0005	no
256/1	adapted	0.0000	[-0.0000, 0.0000]	0.0005	no
256/4	native	-0.0791	[-0.1130, -0.0430]	0.0029	yes
256/4	adapted	-0.0791	[-0.1130, -0.0430]	0.0029	yes
256/16	native	0.4236	[0.2555, 0.6373]	0.0160	yes
256/16	adapted	0.4236	[0.2555, 0.6373]	0.0160	yes
256/31	native	0.8233	[0.5130, 1.2121]	0.0356	yes
256/31	adapted	0.8233	[0.5130, 1.2121]	0.0356	yes
512/1	native	-0.0000	[-0.0000, -0.0000]	0.0073	no
512/1	adapted	-0.0000	[-0.0000, 0.0000]	0.0005	no
512/4	native	-0.0469	[-0.0620, -0.0322]	0.0466	no
512/4	adapted	-0.0797	[-0.1134, -0.0437]	0.0028	yes
512/16	native	-0.0841	[-0.2677, 0.1328]	0.1475	no
512/16	adapted	0.3513	[0.1976, 0.5562]	0.0145	yes
512/31	native	-0.0188	[-0.3506, 0.3434]	0.2071	no
512/31	adapted	0.6828	[0.4036, 1.0544]	0.0288	yes
1024/1	native	0.0000	[-0.0000, 0.0000]	0.0108	no
1024/1	adapted	-0.0000	[-0.0000, 0.0000]	0.0005	no
1024/4	native	-0.0537	[-0.0727, -0.0342]	0.0688	no
1024/4	adapted	-0.0797	[-0.1134, -0.0437]	0.0028	yes
1024/16	native	-0.3506	[-0.5819, -0.0767]	0.2162	no
1024/16	adapted	0.3514	[0.1977, 0.5562]	0.0145	yes
1024/31	native	-0.6128	[-1.1315, -0.0870]	0.3572	no
1024/31	adapted	0.6829	[0.4036, 1.0545]	0.0288	yes